

## **LAB EXPERIMENTS – ITA0403 STATISTICS WITH R PROGRAMMING**

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1. Create numeric, character, and logical vectors and display type and content.

Program:

```
# Numeric vector
```

```
num <- c(10, 20, 30)
```

```
# Character vector
```

```
char <- c("A", "B", "C")
```

```
# Logical vector
```

```
logi <- c(TRUE, FALSE, TRUE)
```

```
print(num)
```

```
print(typeof(num))
```

```
print(char)
```

```
print(typeof(char))
```

```
print(logi)
```

```
print(typeof(logi))
```

output:

```
> # Numeric vector
> num <- c(10, 20, 30)
>
> # Character vector
> char <- c("A", "B", "C")
>
> # Logical vector
> logi <- c(TRUE, FALSE, TRUE)
>
> print(num)
[1] 10 20 30
> print(typeof(num))
[1] "double"
>
> print(char)
[1] "A" "B" "C"
> print(typeof(char))
[1] "character"
>
> print(logi)
[1] TRUE FALSE TRUE
> print(typeof(logi))
[1] "logical"
```

## 2. Create labeled matrices (5×4, 3×3, 2×2) filled by row/column

### Program

# 5×4 matrix (filled by row)

```
m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)
```

```
rownames(m1) <- c("R1","R2","R3","R4","R5")
```

```
colnames(m1) <- c("C1","C2","C3","C4")
```

```
print(m1)
```

# 3×3 matrix (filled by column)

```
m2 <- matrix(1:9, nrow=3, byrow=FALSE)
```

```
rownames(m2) <- c("A","B","C")
```

```
colnames(m2) <- c("X","Y","Z")
```

```
print(m2)
```

# 2×2 matrix

```
m3 <- matrix(1:4, nrow=2, byrow=TRUE)
```

```
rownames(m3) <- c("R1","R2")
```

```
colnames(m3) <- c("C1","C2")
```

```
print(m3)
```

**output:**

```
> # 5x4 matrix (filled by row)
> m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)
> rownames(m1) <- c("R1","R2","R3","R4","R5")
> colnames(m1) <- c("C1","C2","C3","C4")
>
> print(m1)
  C1 C2 C3 C4
R1  1  2  3  4
R2  5  6  7  8
R3  9 10 11 12
R4 13 14 15 16
R5 17 18 19 20
>
> # 3x3 matrix (filled by column)
> m2 <- matrix(1:9, nrow=3, byrow=FALSE)
> rownames(m2) <- c("A","B","C")
> colnames(m2) <- c("X","Y","Z")
>
> print(m2)
  X Y Z
A 1 4 7
B 2 5 8
C 3 6 9
>
> # 2x2 matrix
> m3 <- matrix(1:4, nrow=2, byrow=TRUE)
> rownames(m3) <- c("R1","R2")
> colnames(m3) <- c("C1","C2")
>
> print(m3)
  C1 C2
R1  1  2
R2  3  4
```

### 3. Create and display a 3D array

#### Program

```
arr <- array(1:24, dim=c(2,3,4))  
print(arr)
```

#### output:

```
>  
> arr <- array(1:24, dim=c(2,3,4))  
> print(arr)  
, , 1  
  
      [,1] [,2] [,3]  
[1,]     1     3     5  
[2,]     2     4     6  
  
, , 2  
  
      [,1] [,2] [,3]  
[1,]     7     9    11  
[2,]     8    10    12  
  
, , 3  
  
      [,1] [,2] [,3]  
[1,]    13    15    17  
[2,]    14    16    18  
  
, , 4  
  
      [,1] [,2] [,3]  
[1,]    19    21    23  
[2,]    20    22    24
```

### 4. Create arrays with dimension names and print a specific element

#### Program

```
v <- 1:12
```

```
arr <- array(v,  
             dim=c(2,3,2),  
             dimnames=list(  
               Row=c("R1","R2"),  
               Col=c("C1","C2","C3"),  
               Table=c("T1","T2")
```

))

print(arr)

print(arr["R2","C3","T2"])

### output:

```
>
> v <- 1:12
>
> arr <- array(v,
+             dim=c(2,3,2),
+             dimnames=list(
+                 ROW=c("R1", "R2"),
+                 Col=c("C1", "C2", "C3"),
+                 Table=c("T1", "T2")
+             ))
>
> print(arr)
, , Table = T1

  Col
ROW C1 C2 C3
R1  1  3  5
R2  2  4  6

, , Table = T2

  Col
ROW C1 C2 C3
R1  7  9 11
R2  8 10 12

>
> print(arr["R2","C3","T2"])
[1] 12
```

## 5. Create and manipulate factor variables

### Program

```
heights <- factor(c("Short","Medium","Tall","Medium","Tall"))
```

print(heights)

print(levels(heights))

```
letters_factor <- factor(sample(LETTERS[1:5],10,replace=TRUE))
```

```
print(letters_factor)
```

```
summary(letters_factor)
```

**output:**

```
>
> print(arr["R2", "C3", "T2"])
[1] 12
>
>
> heights <- factor(c("Short", "Medium", "Tall", "Medium", "Tall"))
>
> print(heights)
[1] Short Medium Tall Medium Tall
Levels: Medium Short Tall
> print(levels(heights))
[1] "Medium" "Short" "Tall"
>
> letters_factor <- factor(sample(LETTERS[1:5], 10, replace=TRUE))
>
> print(letters_factor)
[1] C C C C B A C C D A
Levels: A B C D
> summary(letters_factor)
 A B C D
2 1 6 1
```

## 6. Create an R list containing vectors, matrices and functions

### Program

```
vec <- c(1,2,3)
```

```
mat <- matrix(1:4, 2, 2)
```

```
mylist <- list(
  Vector=vec,
  Matrix=mat,
  Function=sum
)
```

```
print(mylist)
```

```
mylist$Function(vec)
```

**output:**

```
>
> vec <- c(1,2,3)
>
> mat <- matrix(1:4,2,2)
>
> mylist <- list(
+   Vector=vec,
+   Matrix=mat,
+   Function=sum
+ )
>
> print(mylist)
$Vector
[1] 1 2 3

$Matrix
      [,1] [,2]
[1,]    1    3
[2,]    2    4

$Function
[1] 650
```

**7. Write R programs for basic tasks: Factors of a number, generate a vector of 10 random integers**

**between -50 and 50, print numbers 1–100 with FizzBuzz logic.**

**(a). Find factors of a number**

**Program**

```
num <- 24
```

```
for(i in 1:num)
```

```
{
```

```
  if(num %% i == 0)
```

```
    print(i)
```

```
}
```

### Output:

```
> num <- 24
>
> for(i in 1:num)
+ {
+   if(num %% i == 0)
+     print(i)
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 6
[1] 8
[1] 12
[1] 24
```

### (b). Generate 10 random integers between -50 and 50

#### Program

```
set.seed(100)
```

```
x <- sample(-50:50,10)
```

```
print(x)
```

#### output:

```
>
> set.seed(100)
>
> x <- sample(-50:50,10)
>
> print(x)
[1] 23 38 27 -28 35 19 -47 4 45 -44
```

### (c). FizzBuzz (1 to 100)

#### Program

```
for(i in 1:100)
```

```
{
```



```
if(i %% 3 == 0 && i %% 5 == 0)
  print("FizzBuzz")
else if(i %% 3 == 0)
  print("Fizz")
else if(i %% 5 == 0)
  print("Buzz")
else
  print(i)
}
```

### Output:

```
> for(i in 1:100)
+ {
+   if(i %% 3 == 0 && i %% 5 == 0)
+     print("FizzBuzz")
+   else if(i %% 3 == 0)
+     print("Fizz")
+   else if(i %% 5 == 0)
+     print("Buzz")
+   else
+     print(i)
+ }
[1] 1
[1] 2
[1] "Fizz"
[1] 4
[1] "Buzz"
[1] "Fizz"
[1] 7
[1] 8
[1] "Fizz"
[1] "Buzz"
[1] 11
[1] "Fizz"
[1] 13
[1] 14
[1] "FizzBuzz"
```